

Tutorial – 01

1. Write a program to print “Hello World”.

Code:

```
using System;
```

```
namespace Practicals
```

```
{
```

```
    internal class T1_01
```

```
    {
```

```
        public static void HelloWorld()
```

```
        {
```

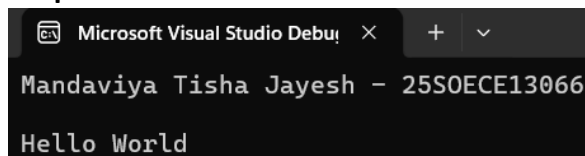
```
            Console.WriteLine("Hello World");
```

```
        }
```

```
    }
```

```
}
```

Output:



2. Design your profile page as given below.

\$\$

Name: your name

DOB: 15/10/1991

Address: 4, xyx society,
Kalawad Road

City: Rajkot

Pincode: 360 001

State: Gujarat

Country: India

Email: abc@ymail.com

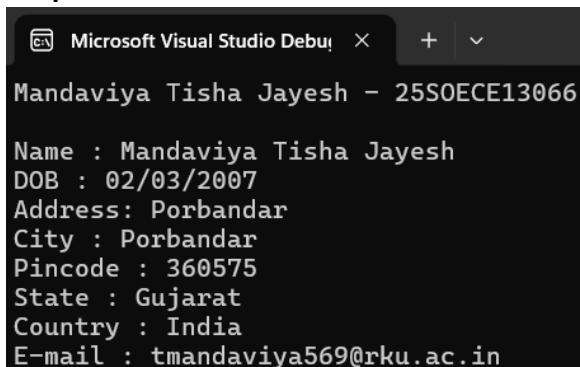
\$\$

Code:

```
using System;
```

```
namespace Practicals
{
    internal class T1_02
    {
        public static void Profile()
        {
            Console.WriteLine("Name: Mandaviya Tisha Jayesh");
            Console.WriteLine("DOB: 02/03/2007");
            Console.WriteLine("Address: Porbandar");
            Console.WriteLine("City: Porbandar");
            Console.WriteLine("Pin Code: 360575");
            Console.WriteLine("State: Gujarat");
            Console.WriteLine("Country: India");
            Console.WriteLine("Email: tmandaviya569@rku.ac.in");
        }
    }
}
```

Output:



Microsoft Visual Studio Debug Console

Mandaviya Tisha Jayesh - 25SOECE13066

Name : Mandaviya Tisha Jayesh
DOB : 02/03/2007
Address: Porbandar
City : Porbandar
Pincode : 360575
State : Gujarat
Country : India
E-mail : tmandaviya569@rku.ac.in

3. Find out whether the given number is odd or even.

Code:

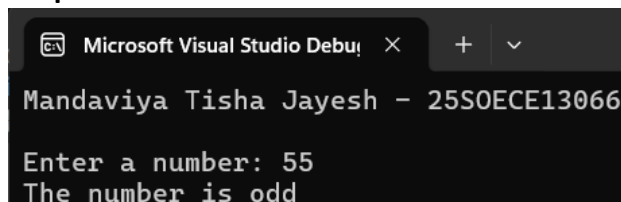
```
using System;
```

```
namespace Practicals
{
    internal class T1_03
    {
        public static void OddEven()
        {
            int number;
```

```
Console.WriteLine("Enter a number:");
number = Convert.ToInt32(Console.ReadLine());

if (number % 2 == 0)
{
    Console.WriteLine("Number is Even.");
}
else
{
    Console.WriteLine("Number is Odd.");
}
}
```

Output:



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Enter a number: 55

The number is odd

4. Rearrange the given code to correct the program. The resultant program will be to input a number and print whether the given number is odd or even.

```
namespace ConsoleApplication1
{
    {
        static void Main(string[] args)
        {
            int x;
            Console.WriteLine("Enter Number : ");
            x = Convert.ToInt32(str);
            Console.WriteLine("Number is Even");
            else
            Console.Read();
            string str = Console.ReadLine();
            if (x % 2 == 0)
            Console.WriteLine("Number is Odd");
        }
    }
}
```

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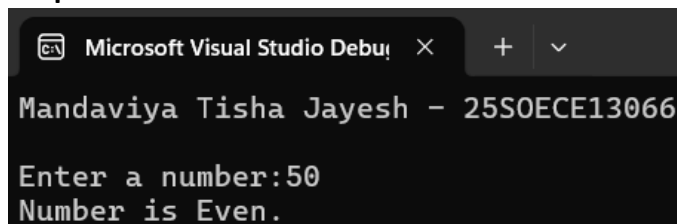
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class Program
using System;
Output:
Enter Number: 10
Number is Even

Code:

```
using System;  
  
namespace Practicals  
{  
    class T1_04  
    {  
        public static void Oddevenstr()  
        {  
            int number;  
            Console.WriteLine("Enter a number:");  
            string str = Console.ReadLine();  
            number = Convert.ToInt32(str);  
  
            if (number % 2 == 0)  
            {  
                Console.WriteLine("Number is Even.");  
            }  
            else  
            {  
                Console.WriteLine("Number is Odd.");  
            }  
        }  
    }  
}
```

Output:



```
Microsoft Visual Studio Debug Console  
Mandaviya Tisha Jayesh - 25SOECE13066  
Enter a number:50  
Number is Even.
```

5. Write output of the program. Also write comment for each line for the following code.

```
using System;
namespace ConsoleApplication1
{
    class Program
    {
        static void Main(string[] args)
        {
            int n,fact=1;
            Console.WriteLine("Enter Number : ");
            string str = Console.ReadLine();
            n = Convert.ToInt32(str);
            for (int i = 1; i <= n; i++)
            {
                fact = fact * i;
            }
            Console.WriteLine("Factorial : {0}",fact);
            Console.Read();
        }
    }
}
```

Code:

```
using System;

namespace Practicals
{
    internal class T1_05
    {
        public static void Factorial()
        {
            Console.WriteLine("Enter a number: ");
            int n = int.Parse(Console.ReadLine());
            int fact = 1;

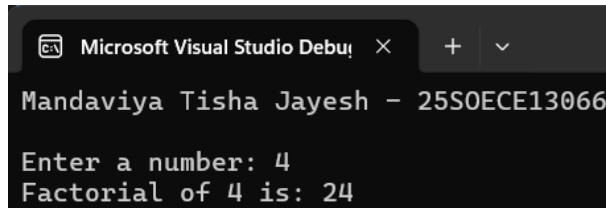
            for (int i = 1; i <= n; i++)
            {
                fact *= i;
            }
        }
    }
}
```

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```
        Console.WriteLine($"Factorial of {n} is {fact}");
    }
}
}
```

Output:



```
Microsoft Visual Studio Debug Console
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Enter a number: 4
Factorial of 4 is: 24
```

6. Write missing statement to get the desired output.

```
using System;
namespace ConsoleApplication1
{
    class Program
    {
        static void Main(string[] args)
        {
            int a,b,c,result;
            Console.Write("Enter Number 1: ");
            //Missing statement
            a = Convert.ToInt32(str);
            Console.Write("Enter Number 2: ");
            //Missing statement
            b = Convert.ToInt32(str);
            Console.Write("Enter Number 3: ");
            str = Console.ReadLine();
            //Missing statement
            result = Sum(a, b, c);
            //Missing statement
            Console.Read();
        }
        static int Sum(int x, int y, int z)
        {
            int res;
            res = x+y+z;
            return res;
        }
    }
}
```

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```
}  
}
```

Output:

```
Enter Number 1 : 10  
Enter Number 2 : 20  
Enter Number 3 : 30  
Sum : 60
```

Code:

```
using System;
```

```
namespace Practicals
```

```
{
```

```
    internal class T1_06
```

```
    {
```

```
        public static void Sumof3()
```

```
        {
```

```
            Console.Write("Enter 1 Number: ");
```

```
            int a = int.Parse(Console.ReadLine());
```

```
            Console.Write("Enter 2 Number: ");
```

```
            int b = int.Parse(Console.ReadLine());
```

```
            Console.Write("Enter 3 Number: ");
```

```
            int c = int.Parse(Console.ReadLine());
```

```
            int ans = a + b + c;
```

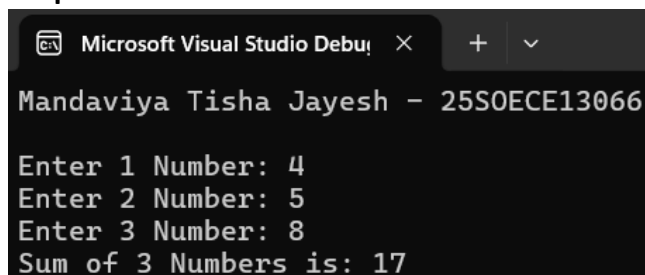
```
            Console.WriteLine("Sum of 3 Number is: " + ans);
```

```
        }
```

```
    }
```

```
}
```

Output:



```
Microsoft Visual Studio Debug Console  
Mandaviya Tisha Jayesh - 25SOECE13066  
Enter 1 Number: 4  
Enter 2 Number: 5  
Enter 3 Number: 8  
Sum of 3 Numbers is: 17
```

7. Predict and write the output of the given code.

```
using System;
namespace While_Loop
{
    class Program
    {
        static void Main(string[] args)
        {
            int num1,res, i;
            Console.WriteLine("Enter a number");
            num1 = Convert.ToInt32(Console.ReadLine());
            i = 1; //Initialization
            //Check whether condition matches or not
            while (i <= 10)
            {
                res = num1 * i;
                Console.WriteLine("{0} x {1} = {2}", num1, i, res);
                i++; //Increment by one
            }
            Console.ReadLine();
        }
    }
}
```

Code:

```
using System;

namespace Practicals
{
    internal class T1_07
    {
        public static void Table()
        {
            int num1, res, i;

            Console.Write("Enter a number: ");
            num1 = Convert.ToInt32(Console.ReadLine());

            i = 1;
            while (i <= 10)
```

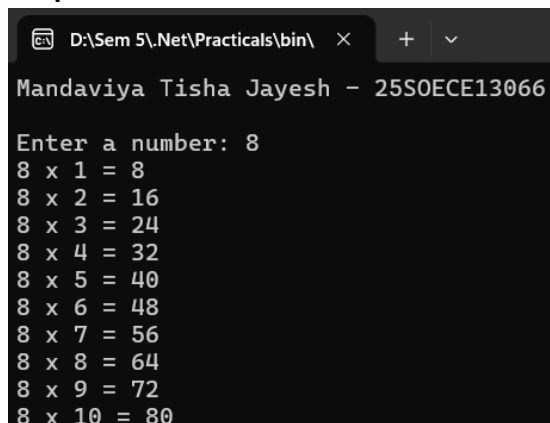

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```
{
    res = num1 * i;
    Console.WriteLine("{0} x {1} = {2}", num1, i, res);

    i++;
}
Console.ReadLine();
}
```

Output:



```
D:\Sem 5\Net\Practicals\bin\
Mandaviya Tisha Jayesh - 25SOECE13066

Enter a number: 8
8 x 1 = 8
8 x 2 = 16
8 x 3 = 24
8 x 4 = 32
8 x 5 = 40
8 x 6 = 48
8 x 7 = 56
8 x 8 = 64
8 x 9 = 72
8 x 10 = 80
```

8. Write a program to convert given name in upper characters.

INPUT : John F Kennedy

OUTPUT: JOHN F KENNEDY

Code:

```
using System;
```

```
namespace Practicals
```

```
{
```

```
    internal class T1_08
```

```
    {
```

```
        public static void Lowertoupper()
```

```
        {
```

```
            string name;
```

```
            Console.Write("Enter name: ");
```

```
            name = Console.ReadLine();
```

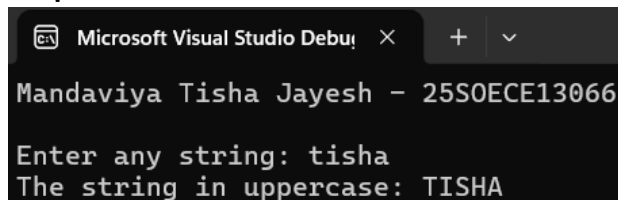
```
            Console.WriteLine("Uppercase: " + name.ToUpper());
```

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```
}  
}  
}
```

Output:



```
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Enter any string: tisha  
The string in uppercase: TISHA
```

9. Write a Program to convert given name in toggle case.

INPUT : JoHn F kEnNedy

OUTPUT: jOhN f KeNneDY

Code:

```
using System;
```

```
namespace Practicals
```

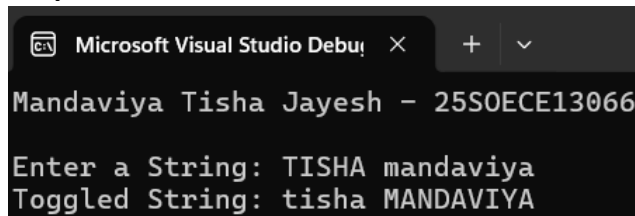
```
{  
    internal class T1_09  
    {  
        public static void ToggleCase()  
        {  
            Console.WriteLine("Enter a String: ");  
            string str = Console.ReadLine();  
            string toggledStr = "";  
            foreach (char c in str)  
            {  
                if (char.IsUpper(c))  
                {  
                    toggledStr += char.ToLower(c);  
                }  
                else if (char.IsLower(c))  
                {  
                    toggledStr += char.ToUpper(c);  
                }  
                else  
                {  
                    toggledStr += c;  
                }  
            }  
        }  
    }  
}
```

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```
    }  
    Console.WriteLine("Toggled String: " + toggledStr);  
  }  
}  
}
```

Output:



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Enter a String: TISHA mandaviya
Toggled String: tisha MANDAVIYA

10. Write a Program which accepts mobile no as a string from the user and converts the last 5 digits into X.

INPUT : 1234567890

OUTPUT: 12345XXXXX

Code:

```
using System;
```

```
namespace Practicals
```

```
{
```

```
    internal class T1_10
```

```
    {
```

```
        public static void MobileNum()
```

```
        {
```

```
            Console.Write("Enter Mobile Number:");
```

```
            string num = Console.ReadLine();
```

```
            if (num.Length >= 5)
```

```
            {
```

```
                string res = num.Substring(0, num.Length - 5) + "XXXXX";
```

```
                Console.WriteLine(res);
```

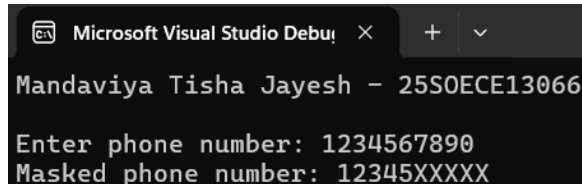
```
            }
```

```
        }
```

```
    }
```

```
}
```

Output:



```
Microsoft Visual Studio Debug Console
Mandaviya Tisha Jayesh - 25SOECE13066
Enter phone number: 1234567890
Masked phone number: 12345XXXXX
```

11. Write a Program which accepts name and gender from the user. Here, gender may have only 1 character, M or F.

Based on the gender prefix the name Mr. & Ms.

NAME : Hillary Clinton

GENDER : F

Code:

```
using System;
```

```
namespace Practicals
```

```
{
```

```
    internal class T1_11
```

```
    {
```

```
        public static void Namegen()
```

```
        {
```

```
            Console.WriteLine("Enter name: ");
```

```
            string name=Console.ReadLine();
```

```
            Console.WriteLine("Enter Gender: ");
```

```
            char gen = Convert.ToChar(Console.ReadLine());
```

```
            if (gen == 'M' || gen == 'm')
```

```
            {
```

```
                Console.WriteLine("Hello Mr. " + name);
```

```
            }
```

```
            else if (gen == 'F' || gen == 'f')
```

```
            {
```

```
                Console.WriteLine("Hello Ms. " + name);
```

```
            }
```

```
            else
```

```
            {
```

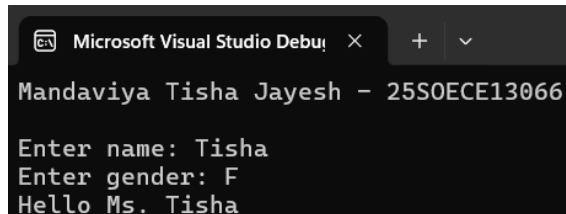
```
                Console.WriteLine("Invalid gender input.");
```

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```
}  
}  
}  
}
```

Output:



```
Microsoft Visual Studio Debug Console  
Mandaviya Tisha Jayesh - 25SOECE13066  
Enter name: Tisha  
Enter gender: F  
Hello Ms. Tisha
```

12. Write a Program which accepts name from the user and prints the same
INPUT : Winston Churchill
OUTPUT: Winston Churchill

Code:

```
using System;
```

```
namespace Practicals
```

```
{
```

```
    internal class T1_12
```

```
    {
```

```
        public static void SameName()
```

```
        {
```

```
            Console.Write("Enter your name: ");
```

```
            string name = Console.ReadLine();
```

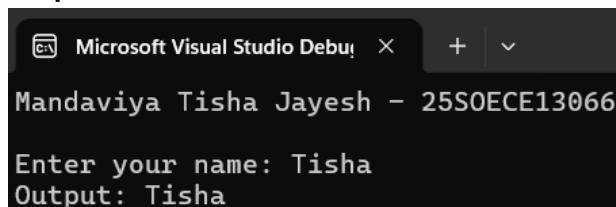
```
            Console.WriteLine("Output: "+name);
```

```
        }
```

```
    }
```

```
}
```

Output:



```
Microsoft Visual Studio Debug Console  
Mandaviya Tisha Jayesh - 25SOECE13066  
Enter your name: Tisha  
Output: Tisha
```

13. Write a Program to prints the following series
0 1 1 2 3 5 8 13 21 34 55

Code:

```
using System;
```

```
namespace Practicals
```

```
{
```

```
    internal class T1_13
```

```
    {
```

```
        public static void Fibonacciseries()
```

```
        {
```

```
            int n1 = 0, n2 = 1, n3;
```

```
            Console.Write(n1 + " " + n2 + " ");
```

```
            {
```

```
                for (int i = 2; i <= 10; i++)
```

```
                {
```

```
                    n3 = n1 + n2;
```

```
                    Console.Write(n3 + " ");
```

```
                    n1 = n2;
```

```
                    n2 = n3;
```

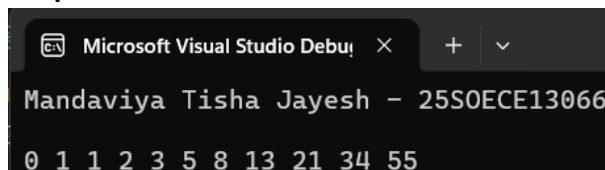
```
                }
```

```
            }
```

```
        }
```

```
    }
```

Output:



```
Microsoft Visual Studio Debug Console
Mandaviya Tisha Jayesh - 25SOECE13066
0 1 1 2 3 5 8 13 21 34 55
```

14. Write a Program which accepts no from the user and print the same in words.
INPUT : 98732
OUTPUT: Nine Eight Seven Three Two

Code:

```
using System;
```

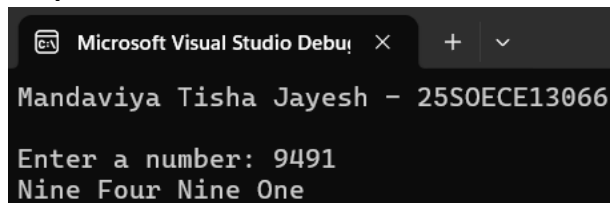
```
namespace Practicals
{
    internal class T1_14
    {
        public static void NumbertoText()
        {
            Console.Write("Enter a number:");
            string num=Console.ReadLine();
            foreach(char c in num)
            {
                switch (c)
                {
                    case '0':
                        Console.WriteLine("Zero");
                        break;
                    case '1':
                        Console.WriteLine("One");
                        break;
                    case '2':
                        Console.WriteLine("Two");
                        break;
                    case '3':
                        Console.WriteLine("Three");
                        break;
                    case '4':
                        Console.WriteLine("Four");
                        break;
                    case '5':
                        Console.WriteLine("Five");
                        break;
                    case '6':
                        Console.WriteLine("Six");
                        break;
                    case '7':
                        Console.WriteLine("Seven");
                        break;
                    case '8':
                        Console.WriteLine("Eight");
                        break;
                    case '9':
```

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```
        Console.WriteLine("Nine");
        break;
    }
}
}
```

Output:



```
Microsoft Visual Studio Debug Console
Mandaviya Tisha Jayesh - 25SOECE13066
Enter a number: 9491
Nine Four Nine One
```

15. Write a Program to check whether the given no is Armstrong no or not.

Code:

```
using System;
```

```
namespace Practicals
```

```
{
```

```
    internal class T1_15
```

```
    {
```

```
        public static void Armstrongnum()
```

```
        {
```

```
            Console.Write("Enter a number:");
```

```
            int num = Convert.ToInt32(Console.ReadLine());
```

```
            int original = num;
```

```
            int rem, arm = 0;
```

```
            while (num != 0)
```

```
            {
```

```
                rem = num % 10;
```

```
                arm = arm + (rem * rem * rem);
```

```
                num = num / 10;
```

```
            }
```

```
            if (original == arm)
```

```
            {
```

```
                Console.WriteLine("This is Armstrong number.");
```

```
            }
```

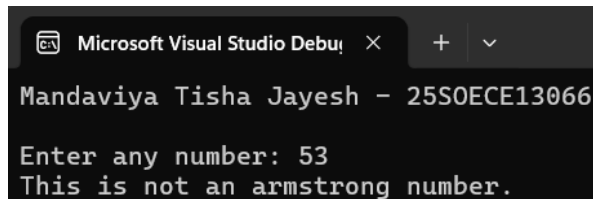
```
        }
    }
}
```


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```
    {  
        Console.WriteLine("This is not Armstrong number.");  
    }  
}  
}
```

Output:



```
Microsoft Visual Studio Debu! X + v  
Mandaviya Tisha Jayesh - 25SOECE13066  
Enter any number: 53  
This is not an armstrong number.
```

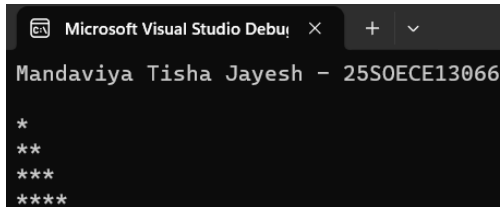
16. Write a program to display a pattern like a right angle triangle using an asterisk
The pattern like :

```
*  
**  
***  
****
```

Code:

```
using System;  
  
namespace Practicals  
{  
    internal class T1_16  
    {  
        public static void Pyramid()  
        {  
            for (int i = 1; i <= 4; i++)  
            {  
                for (int j = 1; j <= i; j++)  
                {  
                    Console.Write("*");  
                }  
                Console.WriteLine();  
            }  
        }  
    }  
}
```

Output:



```
Microsoft Visual Studio Debug Console
Mandaviya Tisha Jayesh - 25SOECE13066
*
**
***
****
```

17. Write a Program to generate following output.

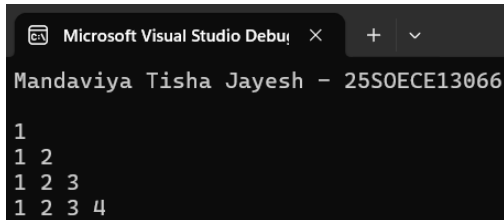
```
1
1 2
1 2 3
1 2 3 4
```

Code:

```
using System;

namespace Practicals
{
    internal class T1_17
    {
        public static void PyramidNum()
        {
            for (int i = 1; i <= 4; i++)
            {
                for (int j = 1; j <= i; j++)
                {
                    Console.Write(j + " ");
                }
                Console.WriteLine();
            }
        }
    }
}
```

Output:



```
Microsoft Visual Studio Debug Console
Mandaviya Tisha Jayesh - 25SOECE13066
1
1 2
1 2 3
1 2 3 4
```

18. Write a program to make such a pattern like a right angle triangle with the number increased by

1.

The pattern like :

```
1
2 3
4 5 6
7 8 9 10
```

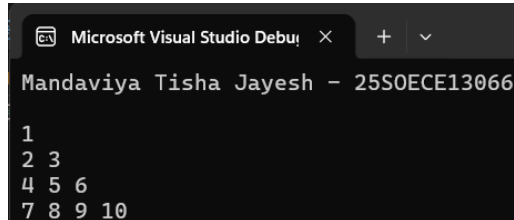
Code:

```
using System;

namespace Practicals
{
    internal class T1_18
    {
        public static void NumberIncreasePyramid()
        {
            int num = 1;

            for (int i = 1; i <= 4; i++)
            {
                for (int j = 1; j <= i; j++)
                {
                    Console.Write(num + " ");
                    num++;
                }
                Console.WriteLine();
            }
        }
    }
}
```

Output:



```
Microsoft Visual Studio Debug Console
Mandaviya Tisha Jayesh - 25SOECE13066
1
2 3
4 5 6
7 8 9 10
```

19. Write a program to make such a pattern as a pyramid with an asterisk.

```
*
* *
* * *
* * * *
```

Code:

```
using System;

namespace Practicals
{
    internal class T1_19
    {
        public static void Asterisk()
        {
            int rows = 4;

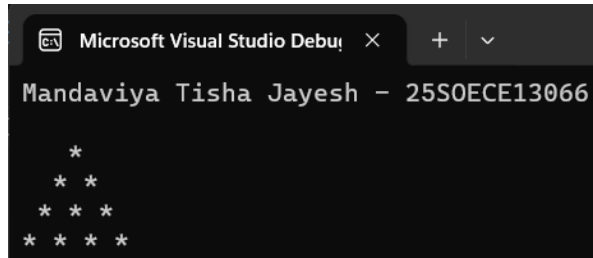
            for (int i = 1; i <= rows; i++)
            {
                for (int j = 1; j <= rows - i; j++)
                {
                    Console.Write(" ");
                }
                for (int k = 1; k <= i; k++)
                {
                    Console.Write("* ");
                }
                Console.WriteLine();
            }
        }
    }
}
```

25SOECE13066

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```
}
```

Output:



```
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  *
 * *
* * *
* * * *
```

20. Write a program to make a pyramid pattern with numbers increased by 1.

```
1
2 3
4 5 6
7 8 9 10
```

Code:

```
using System;

namespace Practicals
{
    internal class T1_20
    {
        public static void PyramidIncrease()
        {
            int rows = 4;
            int num = 1;

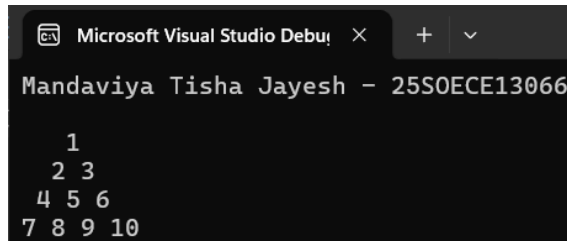
            for (int i = 1; i <= rows; i++)
            {
                for (int j = 1; j <= rows - i; j++)
                {
                    Console.Write(" ");
                }
                for (int k = 1; k <= i; k++)
                {
                    Console.Write(num + " ");
                    num++;
                }
                Console.WriteLine();
            }
        }
    }
}
```

25SOECE13066

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```
}  
}  
}  
}
```

Output:



```
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1  
2 3  
4 5 6  
7 8 9 10
```

21. Write a program to find the sum of the series $5 + 55 + 555 + 5555 + \dots$ n terms.

Test Data :

Input the number of terms : 4

Input number : 5

Expected Output :

5 + 55 + 555 + 5555

The Sum is : 6170

Code:

```
using System;
```

```
namespace Practicals
```

```
{
```

```
    internal class T1_21
```

```
    {
```

```
        public static void T1_21Main()
```

```
        {
```

```
            Console.WriteLine("Enter terms:");
```

```
            int terms = Convert.ToInt32(Console.ReadLine());
```

```
            Console.WriteLine("Enter input number:");
```

```
            int number = Convert.ToInt32(Console.ReadLine());
```

```
            int sum = 0;
```

```
            int current = 0;
```

```
            Console.WriteLine("The series is:");
```

```
            for (int i = 1; i <= terms; i++)
```

25SOECE13066

Enterprise Computing Through .NET Framework (CE525)

```
{
    current = current * 10 + number;
    Console.Write(current);
    sum += current;

    if (i < terms)
    {
        Console.Write(" + ");
    }
}

Console.WriteLine("\n\nSum is " + sum);
}
```

Output:



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Enter terms: 4
Enter input number: 5

5 + 55 + 555 + 5555
Sum is: 6170

22. Write a program to display a pattern like a diamond.

```
*
***
*****
*****
*****
*****
*****
***
*
```

Code:

`using System;`

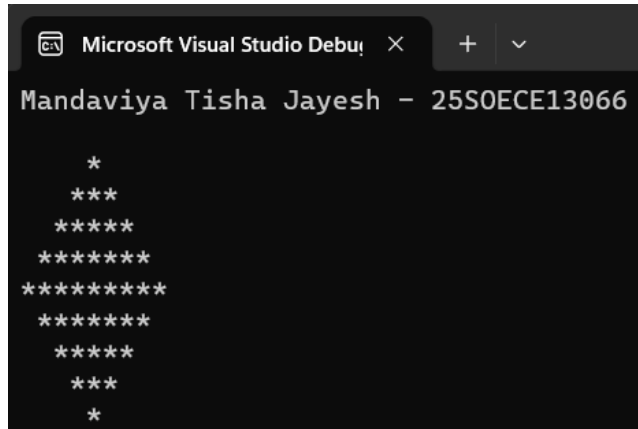
`namespace Practicals`

```
{
    internal class T1_22
    {
        public static void T1_22Main()
        {
            int rows = 5;
            for (int i = 1; i <= rows; i++)
            {
                for (int j = i; j < rows; j++)
                {
                    Console.Write(" ");
                }
                for (int k = 1; k <= (2 * i - 1); k++)
                {
                    Console.Write("*");
                }
                Console.WriteLine();
            }
            for (int i = rows - 1; i >= 1; i--)
            {
                for (int j = rows; j > i; j--)
                {
                    Console.Write(" ");
                }
                for (int k = 1; k <= (2 * i - 1); k++)
                {
                    Console.Write("*");
                }
                Console.WriteLine();
            }
        }
    }
}
```


25SOECE13066

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Output:



```
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  *
 ***
*****
*****
*****
*****
*****
  ***
  *
```